

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

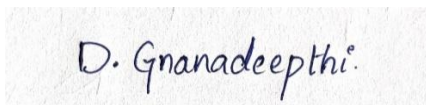
Total Marks: 50

Annexure A

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

a. Compute n $\phi(n)$

- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- a. Encrypt using ECB and CBC modes (Assume IV = 0000)
- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems
- e. Estimate which consumes less power and memory
- f. Justify which cryptosystem is better for the given scenario with reasoning

ANSWERS

2. Block Cipher Mode Evaluation

Given:

Plaintext blocks:

P1 = 1010

P2 = 1010

P3 = 1111

P4 = 1010

Key: K=1100

IV (Initialization Vector): 0000

Assumption: Encryption is performed using XOR with the key.

$$C = P \oplus K$$

a. Encrypt using ECB and CBC modes

ECB (Electronic Codebook) Mode

Encryption:

$$C_i = P_i \oplus K$$

Plaintext	Key	Ciphertext
1010	1100	0110
1010	1100	0110
1111	1100	0011
1010	1100	0110

ECB Ciphertext:

[0110,0110,0011,0110]

CBC (Cipher Block Chaining) Mode

Formula: $C_i = (P_i \oplus C_{i-1}) \oplus K$

Where

$C_0 = IV = 0000$

Block 1

$1010 \oplus 0000 = 1010$

$1010 \oplus 1100 = 0110$

$C_1 = 0110$

Block 2

$1010 \oplus 0110 = 1100$

$1100 \oplus 1100 = 0000$

$C_2 = 0000$

Block 3

$1111 \oplus 0000 = 1111$

$1111 \oplus 1100 = 0011$

$C_3 = 0011$

Block 4

$1010 \oplus 0011 = 1001$

$1001 \oplus 1100 = 0101$

C4 = 0101

CBC Ciphertext:

[0110,0000,0011,0101]

b. Compare the Ciphertext Outputs

Block	Plaintext	ECB	CBC
1	1010	0110	0110
2	1010	0110	0000
3	1111	0011	0011
4	1010	0110	0101

- In ECB, identical plaintext blocks produce identical ciphertext blocks.
- In CBC, identical plaintext blocks produce different ciphertext blocks because each block depends on the previous ciphertext block.

c. Analysis: Which mode hides patterns better?

CBC mode hides patterns better than ECB mode.

Reason:

- In ECB, every identical plaintext block encrypts to the same ciphertext, making repeated patterns visible to an attacker.
- In CBC, each plaintext block is first XORed with the previous ciphertext block (or IV for the first block), so identical plaintext blocks generate different ciphertexts.
- This chaining process makes CBC more secure against pattern analysis and is preferred over ECB for encrypting multiple blocks.

3. DES vs 3-DES vs Blowfish Performance Study

Given:

Algorithm	Key Size	Time per Block
DES	56 bits	2 ms/block
3-DES	168 bits	6 ms/block
Blowfish	128 bits	1 ms/block

Number of blocks = **1000**

i. Calculate Total Encryption Time

Formula:

$$\text{Total Time} = \text{Time per Block} \times \text{Number of Blocks}$$

DES

$$2 \times 1000 = 2000 \text{ ms} = 2 \text{ seconds}$$

3-DES

$$6 \times 1000 = 6000 \text{ ms} = 6 \text{ seconds}$$

Blowfish

$$1 \times 1000 = 1000 \text{ ms} = 1 \text{ second}$$

Results

Algorithm Time per Block Total Time (1000 Blocks)

DES	2 ms	2000 ms (2 s)
3-DES	6 ms	6000 ms (6 s)
Blowfish	1 ms	1000 ms (1 s)

ii. Analyze Trade-offs Between Security and Performance

Algorithm Security		Performance Analysis	
DES	Low (56-bit key is vulnerable to brute-force attacks)	Fast	Good performance but no longer considered secure.
3-DES	High (168-bit key)	Slow	Much stronger security than DES but approximately three times slower.

Algorithm Security

Performance Analysis

Blowfish **High (128-bit key)**

Very Fast

Provides strong security with the best encryption speed among the three.

Analysis:

- DES offers fast encryption but is insecure due to its short 56-bit key.
- 3-DES significantly improves security by applying DES three times, but this increases encryption time.
- Blowfish achieves both strong security and high performance, making it efficient for large amounts of data.

iii. Recommendation for Secure Real-Time Communication

Recommended Algorithm: Blowfish

Reasons:

- Fastest encryption (**1 second for 1000 blocks**).
- Strong **128-bit key** provides good security.
- Suitable for applications requiring **low latency**, such as secure real-time communication.
- Offers a better balance between **security and performance** than DES and 3-DES

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

strong

justification

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
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Accuracy of Calculation	20 %	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15 %	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation